



Bahir Dar University
"Wisdom at the Source of the Blue Nile"

Bahir-Dar Institutions of Technology [BIT] Faculty of Computing
Software Engineering Department Operating System and System
Programming Individual Project Work

Installation of Ubuntu Linux on Oracle VirtualBox



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1.Introduction

Operating systems are essential software that manage computer hardware and software resources. Virtualization technology allows multiple operating systems to run on one physical computer using virtual machines. In this project, Ubuntu Linux operating system was installed on Oracle VirtualBox virtual environment by following the provided tutorial video. Virtualization provides a safe and flexible environment for learning, testing, and development without affecting the host operating system.

VirtualBox allows users to create virtual machines and install guest operating systems virtually. Ubuntu Linux is selected because it is stable, open-source, beginner-friendly, and widely used in education and industry.

2.Objectives

General Objective:

To install Ubuntu Linux operating system on Oracle VirtualBox.

Specific Objectives:

- To understand virtualization technology.
- To create a virtual machine.
- To install Ubuntu Linux successfully.

- To understand Linux filesystem support.
- To identify installation challenges and solutions.

3. Requirements

3.1 Hardware Requirements

	HARDWARE	REQUIREMENTS
1	Processor	Intel/AMD processor
2	RAM	Minimum 4GB
3	Storage	At least 25 freespace
4	Internet	Required for download

Table 1: Hardware specification of the host computer

3.2 Software Requirements:

To download VirtualBox and the Ubuntu ISO file, you only need a modern computer running Windows, macOS, or Linux, along with any up-to-date web browser and a stable

internet connection. Once downloaded, installing and running VirtualBox requires a 64-bit host operating system (Windows 10/11, recent macOS, or Linux), enabled hardware virtualization (Intel VT-x or AMD-V) in the BIOS, and administrator privileges. For smooth performance, it is recommended to have at least 8 GB of RAM (16 GB preferred), a multi-core processor, and sufficient free storage space (30 GB or more) on your host computer. Inside VirtualBox, you should allocate at least 4 GB RAM and 25 GB disk space to the Ubuntu virtual machine.



Figure 1: Downloaded VirtualBox and Ubuntu ISO files

4. Installation Steps

Step 1: Download Oracle VirtualBox

The Oracle VirtualBox virtualization software was downloaded from the official website and installed on the host operating system.

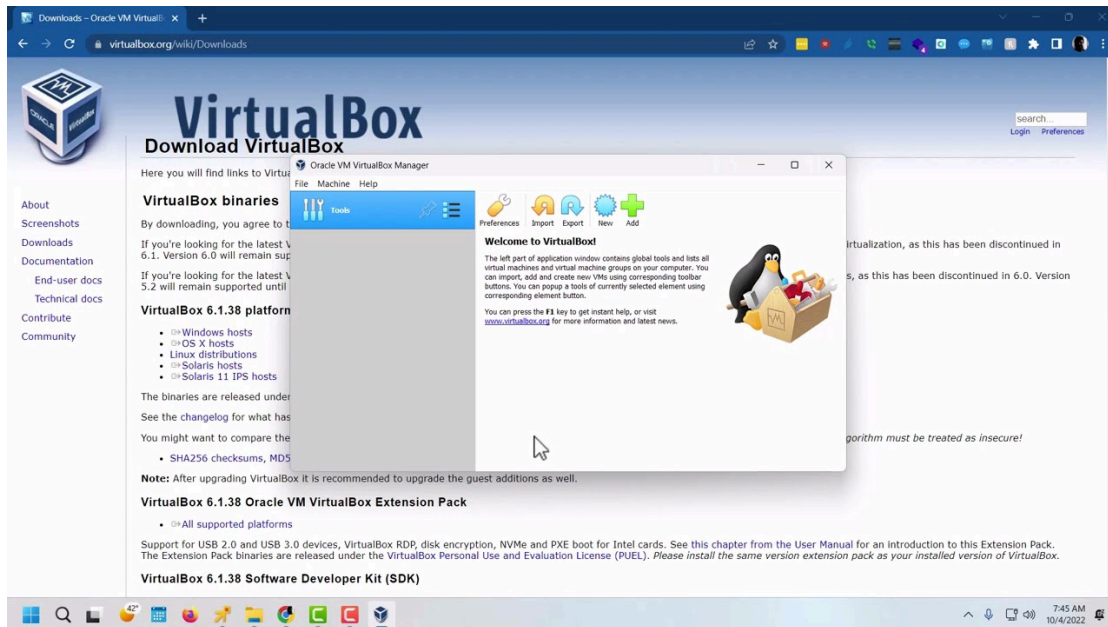


Figure 2: Downloading Oracle VirtualBox

Step 2: Download Ubuntu ISO File

Ubuntu 22.04 LTS ISO image file was downloaded from the official Ubuntu website for installation inside the virtual machine. Ubuntu LTS versions provide long-term support and stability.

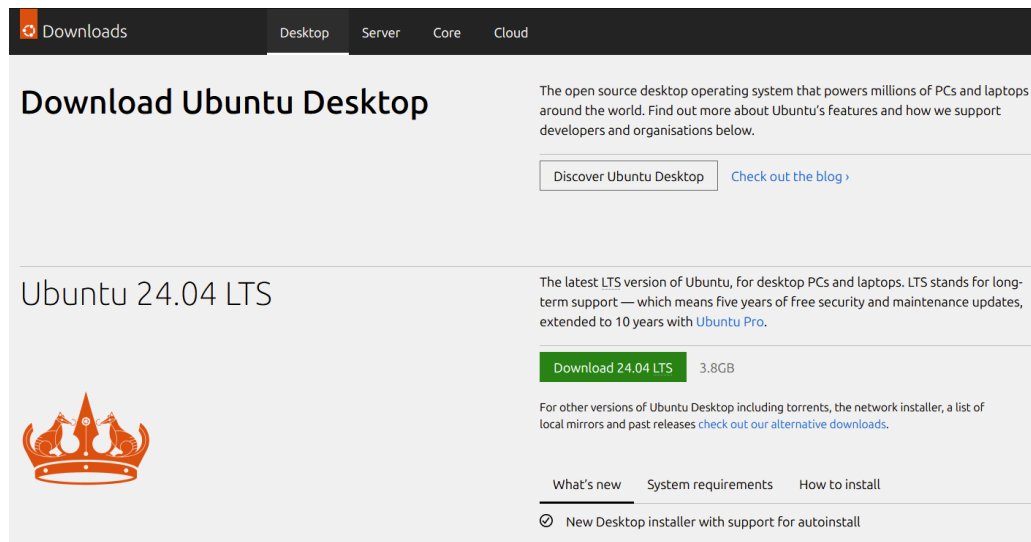


Figure 3: Downloading Ubuntu ISO file

Step 3: Open VirtualBox and Create New VM

After opening VirtualBox, a new virtual machine was created by selecting “New”. The virtual machine name was entered using the student’s full name as instructed.

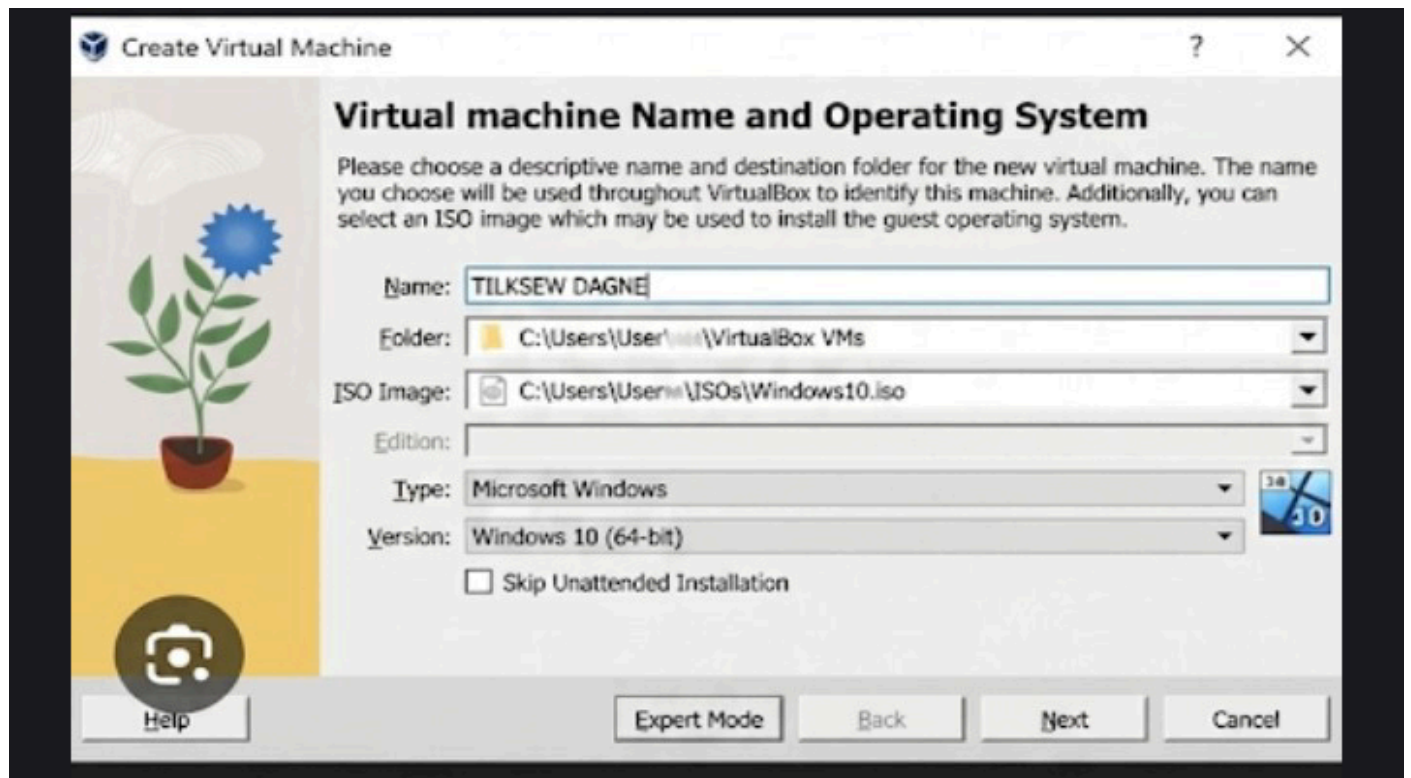


Figure 4: Creating a new virtual machine

Step 4: Configure VM Settings

What to Write

The following configurations were applied:

Type: Linux

Version: Ubuntu (64-bit)

RAM allocation: 4096 MB

Storage allocation: 25 GB

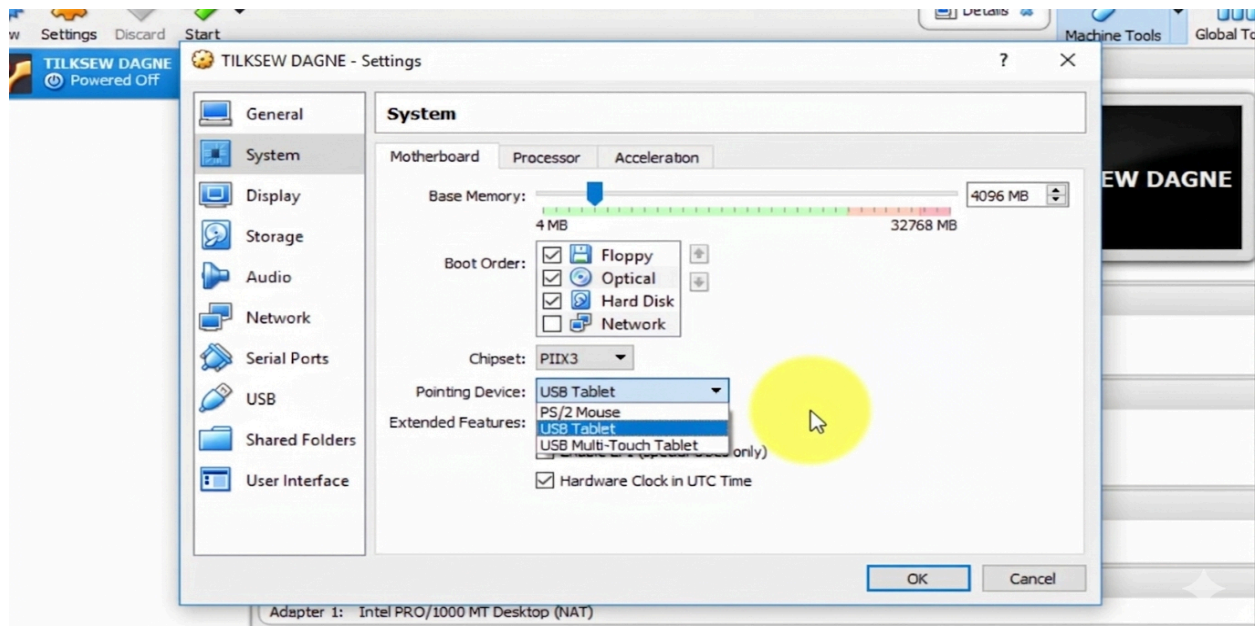


Figure 5: Configuring virtual machine settings

Step 5: Attach Ubuntu ISO File

The Ubuntu ISO image file was attached as a virtual optical disk for booting the operating system installation.

Step 6: Start the Virtual Machine

The virtual machine was successfully powered on in Oracle VirtualBox to initiate the Ubuntu installation process. After attaching the Ubuntu Desktop ISO file and configuring the boot order, the VM booted directly from the ISO image, loading the Ubuntu installer. The boot menu appeared, allowing the selection of "Try or Install Ubuntu," which started the installation wizard inside the virtual environment.

Step 7: Install Ubuntu Linux

Ubuntu installation was started by selecting “Install Ubuntu.” Language, keyboard layout, installation type, and user account information were configured during installation.

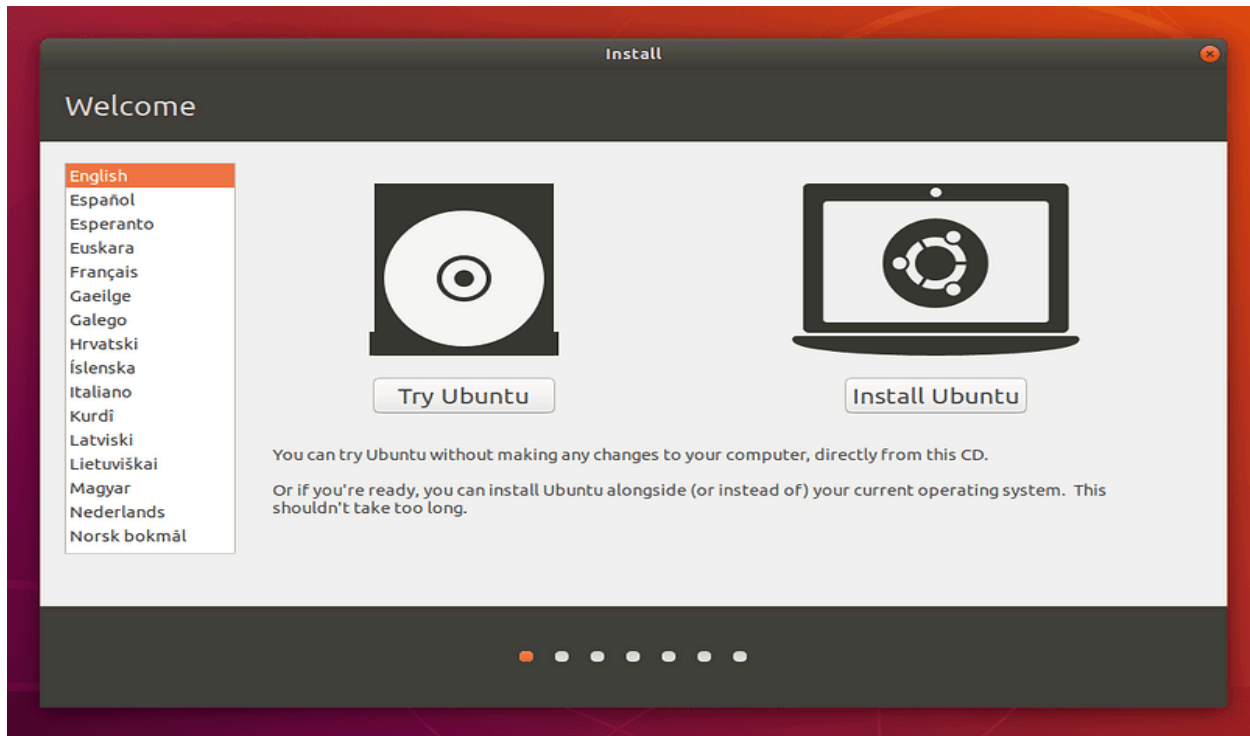
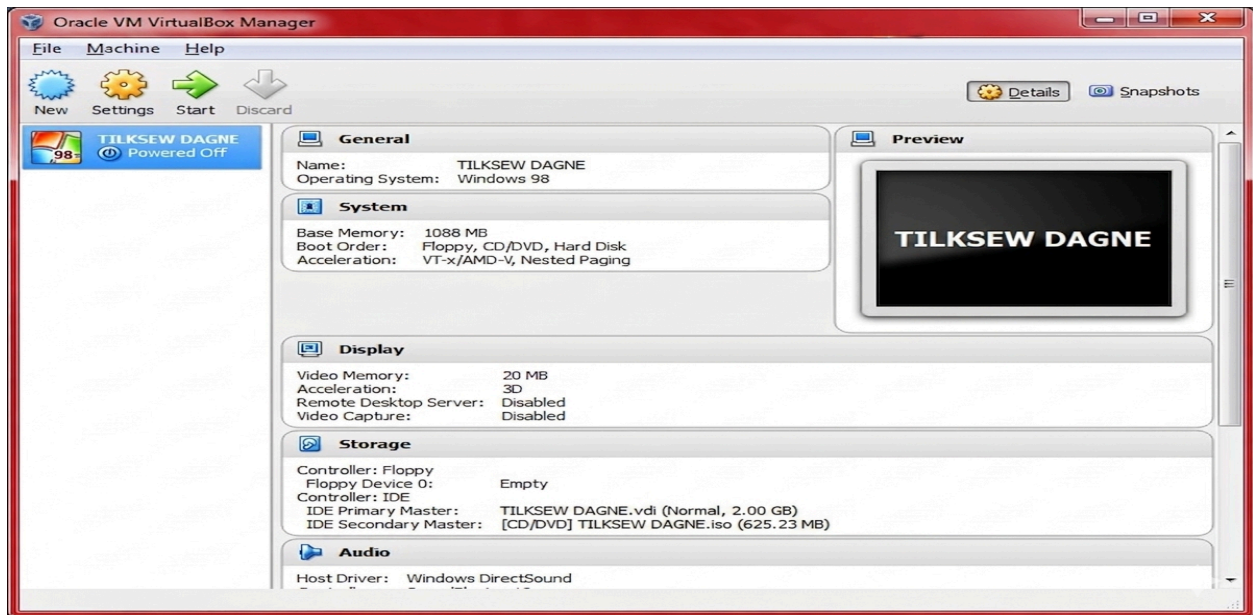


Figure 6: Ubuntu installation welcome screen

Step 8: Complete Installation

After installation completed successfully, the virtual machine was restarted and Ubuntu desktop environment appeared.



Caption: Figure 7: Ubuntu desktop after successful installation

5.Problems Faced

During the Ubuntu installation process in VirtualBox, several common issues may arise that can interrupt or slow down the setup. These include slow virtual machine performance due to insufficient resource allocation, the Ubuntu ISO not being detected by the virtual machine, and virtualization technology (VT-x or AMD-V) being disabled in the system BIOS. Each problem was effectively resolved by increasing the RAM and CPU allocation for better performance, reattaching the ISO image in the VirtualBox settings, and enabling hardware virtualization directly from the BIOS. These troubleshooting steps provided practical experience in diagnosing and fixing typical virtualization-related issues.

Filesystem Support:

Ubuntu Linux primarily utilizes the ext4 filesystem, which offers several key advantages including journaling support for data recovery, faster read/write performance, large file handling capability, and high overall stability. In comparison, other common filesystems such as NTFS (used by Windows), FAT32 and exFAT (commonly used for USB storage and flash drives), and APFS (used by macOS) serve different platforms and purposes. The use of ext4 during the Ubuntu installation ensured a reliable, secure, and efficient storage environment. Screenshots of the disk partitioning and filesystem configuration were captured to document the successful setup.

8. Advantages and Disadvantages

Advantages:

Safe testing environment

No need for dual boot

Easy backup and snapshot support

Cost-effective

Disadvantages:

Uses more RAM

Slower than native installation

Depends on hardware virtualization support.

9. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is the process of creating virtual versions of computer systems using software called hypervisors. Virtual machines allow multiple operating systems to run on one physical computer.

Why Virtualization is Important?

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- Reduces hardware cost
- Improves resource utilization
- Provides safe testing environments
- Supports software development and education

How Virtualization Works?

VirtualBox acts as a Type-2 hypervisor. It allocates CPU, RAM, storage, and network resources to the guest operating system.

Glasp +1

 Screenshot:  Screenshot of running VM inside Windows.

Caption: Figure 15: Ubuntu running virtually on Windows host OS

10. Conclusion

The project successfully demonstrated the complete installation of the Ubuntu Linux operating system within Oracle VirtualBox, a popular open-source virtualization platform. By setting up a virtual machine from scratch, the project provided valuable hands-on experience in key areas such as virtualization concepts, virtual hardware configuration, virtual machine creation and management, disk partitioning, filesystem support, and the step-by-step procedures for installing a Linux distribution.

This practical exercise highlighted the power and flexibility of virtualization technology, which allows multiple operating systems to run simultaneously on a single physical computer in isolated environments. It also helped in developing essential skills including ISO image handling, BIOS/UEFI settings configuration (such as enabling VT-x or AMD-V), and post-installation system optimization.

Overall, virtualization using tools like VirtualBox creates a safe, secure, and risk-free learning platform. It enables users to experiment, test new software, and explore different operating systems without any fear of damaging or altering the host computer's primary setup. This approach is highly beneficial for students, developers, and IT enthusiasts who wish to enhance their technical knowledge efficiently and cost-effectively.

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11.Future Outlook / Recommendation

Future Improvements:

- The following enhancements can be considered for future projects:
- Installing and configuring multiple operating systems (e.g., Windows, Ubuntu, and Fedora) within the same virtualization environment.
- Learning and practicing Linux system administration tasks, including user management, package installation, networking, and security configurations.
- Exploring cloud-based virtualization platforms such as AWS EC2, Microsoft Azure, and Google Cloud.
- Working with advanced virtualization and containerization technologies, including VMware Workstation and Docker.

Recommendation:

Students are strongly encouraged to actively practice virtualization technologies. Regular hands-on experience with tools like VirtualBox, VMware, and Docker will significantly improve their practical understanding of operating systems, enhance technical skills, and better prepare them for real-world IT and software development environments.

12.References

1. Oracle VM VirtualBox Official Documentation. <https://www.virtualbox.org/manual/>
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4. Oracle VM VirtualBox User Manual – Configuring Virtual Machines.
5. Microsoft Documentation. Intel VT-x and AMD-V Virtualization Technology.